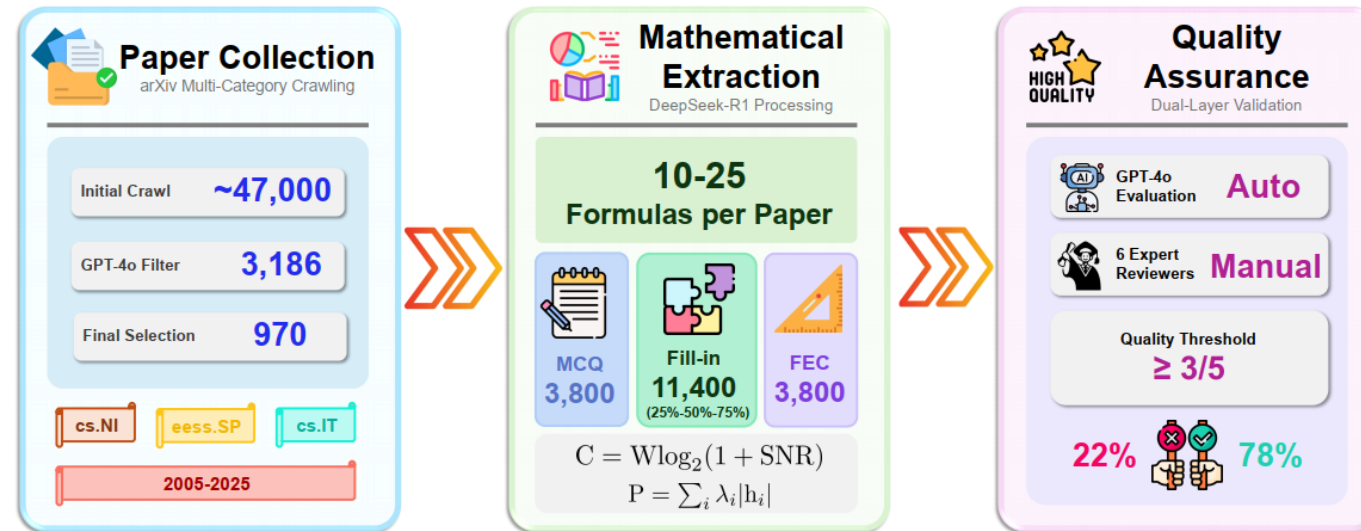


WirelessMathLM: Teaching Mathematical Reasoning for LLMs in Wireless Communications with Reinforcement Learning



From Xin Li, etc.

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WirelessMathBench: A Mathematical Modeling Benchmark for LLMs in Wireless Communications

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Abstract

Large Language Models (LLMs) have achieved impressive results across a broad array of tasks, yet their capacity for complex, domain-specific mathematical reasoning—particularly in wireless communications—remains underexplored. In this work, we introduce WirelessMathBench, a novel benchmark specifically designed to evaluate LLMs on mathematical modeling challenges to wireless communications engineering. Our benchmark consists of 587 meticulously curated questions sourced from 40 state-of-the-art research papers, encompassing a diverse spectrum of tasks ranging from basic multiple-choice questions to complex equation completion tasks, including both partial and full completions, all of which rigorously adhere to physical and dimensional constraints. Through extensive experimentation with leading LLMs, we observe that while many models excel in basic recall tasks, their performance degrades significantly when reconstructing partially or fully obscured equations, exposing fundamental limitations in current LLMs. Even DeepSeek-R1, the best performer on our benchmark, achieves an average accuracy of only 38.05%, with a mere 7.83% success rate in full equation completion. By publicly releasing WirelessMathBench along with the evaluation toolkit, we aim to advance the development of more robust, domain-aware LLMs for wireless system analysis and broader engineering applications.

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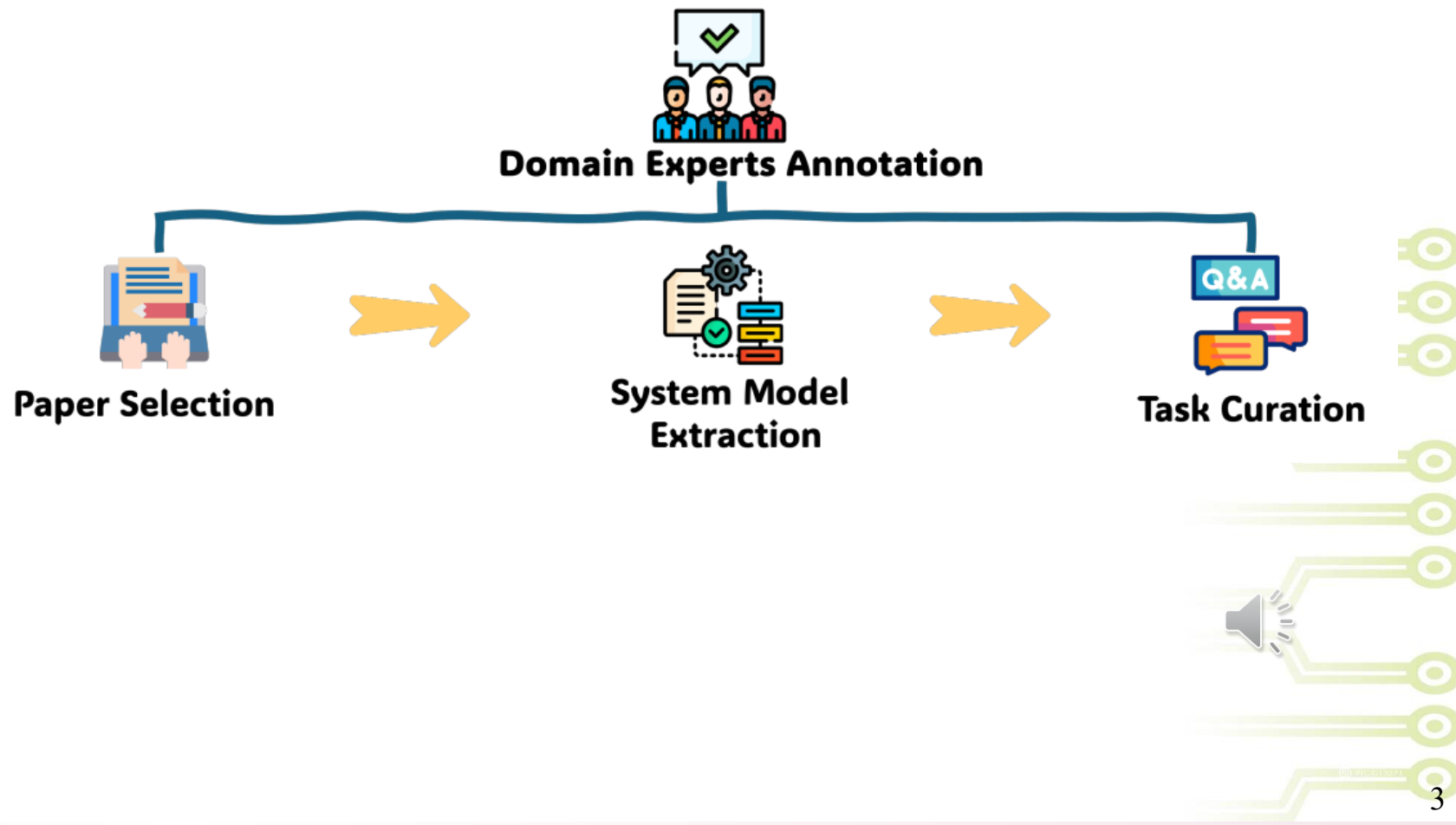
Address: Vienna, Austria

<https://lixin.ai/WirelessMathLM/>



介绍框架

- 背景/简要介绍
- 研究方法
- 实验结果
- 总结



无线大语言模型

• 总体表现

- 制作了无线领域的小型问答数据集 (核心贡献)
- 利用无线数据集对已有的开源大语言模型进行微调
- 微调后的小参数模型(7B)在无线领域专业问题回答的表现上与GPU-4o等更大规模的模型接近

[1] Li, Xin, et al. "WirelessMathLM: Teaching Mathematical Reasoning for LLMs in Wireless Communications with Reinforcement Learning." *arXiv preprint* (2025).

Background

In a double-RIS-assisted massive MIMO system, the overall channel from user k , denoted $\mathbf{h}_k$, incorporates single-reflection links via RIS₁ and RIS₂, as well as a double-reflection link through both RISs. Here, $\mathbf{N}_j$ represents the channel from RIS_j to the BS, and $\mathbf{D}$ denotes the channel between the RISs. The reflection coefficients of RIS_j are given by θ_j .

Multiple Choice Question

$$\mathbf{h}_k = [\text{MASK}]$$

Question: Which expression correctly represents the user- k effective channel at the base station?

Options:



- (A) $\mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \theta_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$.
- (B) $\mathbf{N}_2 \mathbf{h}_{k2} \theta_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{D} \text{diag}(\theta_2) \theta_1$
- (C) $\mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \theta_2 + \mathbf{N}_1 \mathbf{h}_{k1} \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$
- (D) $\mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \theta_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$

Answer: The correct expression is [A].

Progressive Masking Fill-in-the-blank (Level 1, 2, 3)

- L1 $\mathbf{h}_k = \mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) [\text{MASK}] + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$
Question 1: Which reflection vector is missing in the first single-reflection term?
- L2 $\mathbf{h}_k = [\text{MASK}] \text{diag}(\mathbf{h}_{k2}) [\text{MASK}] + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$
Question 2: Fill in the two interdependent channel-related variables for the RIS₂ path.
- L3 $\mathbf{h}_k = [\text{MASK}] \text{diag}(\mathbf{h}_{k2}) [\text{MASK}] + [\text{MASK}] \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$
Question 3: Complete the three missing terms describing the single-reflection link via RIS₁.
- Mask Ratio Answer: $\mathbf{h}_k = \mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \theta_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$

Full Equation Completion

$$\mathbf{h}_k = [\text{MASK}]$$

Question: Write the full expression for the overall effective channel.

Answer: $\mathbf{h}_k = \mathbf{N}_2 \text{diag}(\mathbf{h}_{k2}) \theta_2 + \mathbf{N}_1 \text{diag}(\mathbf{h}_{k1}) \theta_1 + \mathbf{N}_2 \text{diag}(\theta_2) \mathbf{D} \text{diag}(\theta_1) \mathbf{h}_{k1}$

无线大语言模型

数据集制作

- 从arXiv收集并挑选经典无线领域论文的latex版本
 - GPT初筛 + 二次人工选择
 - 970篇各个细分方向的论文
- 从文章中提取出其中的关键公式与对应的背景
 - 系统模型, 变量定义, 假设
 - DeepSeek + Prompt
 - 每篇10-25个公式
- 将提取的内容转换成三种问答题
 - 多选题 / 填空题 / 写公式
 - GPT给出最初回答 + 人工验证

论文背景

关键假设

变量定义

关键公式
与描述

SIM-Based Air-Ground Communication System

Background This paper proposes a Digital Twin (DT)-assisted framework for joint optimization of Stacked Intelligent Metasurface (SIM)-based air-ground communication and electric Vertical Take-off and Landing (eVTOL) flight control within prescribed air corridors. The system model integrates a multi-layer SIM beamforming structure at the Air Traffic Control (ATCo) station with a composite potential field method for eVTOL trajectory planning, aiming to maximize the sum transmission rate while ensuring safe navigation.

Key Assumptions • The air-ground channel between the SIM and each eVTOL follows a Rician fading model.

• Each meta-atom on the SIM imposes an ideal, configurable phase shift without amplitude attenuation.

• The transmission matrices $\mathbf{W}^l$ between metasurface layers are modeled based on Rayleigh-Sommerfeld diffraction theory, assuming perfect knowledge of the SIM's physical structure.

• eVTOLs fly within a predefined, non-overlapping air corridor $\mathcal{R}_{cor}$.

• The signals for different eVTOLs are independent and identically distributed (i.i.d.) with zero mean and unit variance.

• The additive receiver noise is independent, circularly symmetric Gaussian (AWGN).

Parameter Definitions $\mathbf{B} = [x^{ATCo}, y^{ATCo}, 0]^T$ (Type: ATCo station position; Domain: $\mathbb{R}^{3 \times 1}$; Unit: m)

M (Type: Number of eVTOLs / ATCo antennas; Domain: $\mathbb{Z}^+$; Unit: None)

L (Type: Number of metasurface layers in SIM; Domain: $\mathbb{Z}^+$; Unit: None)

K (Type: Number of meta-atoms per metasurface layer; Domain: $\mathbb{Z}^+$; Unit: None)

N (Type: Number of discrete time slots; Domain: $\mathbb{Z}^+$; Unit: None)

δ (Type: Duration of a time slot; Domain: $\mathbb{R}^+$; Unit: s)

$\mathbf{q}_m[n] = [x_m^{eVTOL}[n], y_m^{eVTOL}[n], z_m^{eVTOL}[n]]^T$ (Type: 3D position of eVTOL m at time n ; Domain: $\mathcal{R}_{cor} \subset \mathbb{R}^3$; Unit: m)

v_{max} (Type: Maximum eVTOL velocity; Domain: $\mathbb{R}^+$; Unit: m/s)

P_{ATCo} (Type: Total available transmission power at ATCo; Domain: $\mathbb{R}^+$; Unit: W)

$p_m[n]$ (Type: Transmission power allocated to eVTOL m at time n ; Domain: $\mathbb{R}^+$; Unit: W)

$\theta_k^l[n]$ (Type: Phase shift of meta-atom k on layer l at time n ; Domain: $[0, 2\pi)$; Unit: rad)

$\Psi^l[n] = \text{diag}(e^{j\theta_1^l[n]}, \dots, e^{j\theta_K^l[n]})$ (Type: Phase shift matrix for layer l at time n ; Domain: $\mathbb{C}^{K \times K}$; Unit: None)

$\mathbf{W}^l$ (Type: Transmission matrix between layers $l-1$ and l ; Domain: $\mathbb{C}^{K \times K}$; Unit: None)

$\mathbf{w}_m^1$ (Type: Transmission vector from ATCo antenna m to first metasurface layer; Domain: $\mathbb{C}^{K \times 1}$; Unit: None)

λ (Type: Carrier wavelength; Domain: $\mathbb{R}^+$; Unit: m)

d_x, d_y (Type: Size of a meta-atom along x and y axes; Domain: $\mathbb{R}^+$; Unit: m)

$\mathbf{h}_m^H[n]$ (Type: Channel vector from last SIM layer to eVTOL m at time n ; Domain: $\mathbb{C}^{1 \times K}$; Unit: None)

ρ_0 (Type: Reference path loss at 1m; Domain: $\mathbb{R}^+$; Unit: None (often in dB))

α^h (Type: Path loss exponent; Domain: $\mathbb{R}_{\geq 2}$; Unit: None)

κ^h (Type: Rician factor; Domain: $\mathbb{R}^+$; Unit: dB) σ_m^2 (Type: Receiver noise power at eVTOL m ; Domain: $\mathbb{R}^+$; Unit: W)

$s_m[n]$ (Type: Transmission data symbol for eVTOL m at time n ; Domain: $\mathbb{C}$; Unit: None; Assumption: $\mathbb{E}\{s_m[n]\} = 0$, $\mathbb{E}\{|s_m[n]|^2\} = 1$, i.i.d.)

Core Equations 1) SIM Beamforming Matrix.

The end-to-end beamforming matrix $\mathbf{G}[n]$ of the L -layer SIM is given by the product of the transmission and phase shift matrices across all layers.

$$\mathbf{G}[n] = \Psi^L[n] \mathbf{W}^L \Psi^{L-1}[n] \dots \Psi^2[n] \mathbf{W}^2 \Psi^1[n] \in \mathbb{C}^{K \times K}$$

2) Inter-layer Transmission Matrix Entry.

The (k, k') -th entry of the transmission matrix $\mathbf{W}^l$ is derived from Rayleigh-Sommerfeld diffraction theory.

$$w_{k,k'}^l = \frac{d_x d_y \cos \chi_{k,k'}^l}{d_{k,k'}^l} \left(\frac{1}{2\pi d_{k,k'}^l} - j \frac{1}{\lambda} \right) e^{j2\pi d_{k,k'}^l / \lambda}$$

where $d_{k,k'}^l$ is the distance between meta-atoms, and $\chi_{k,k'}^l$ is the angle between the propagation direction and the normal to the layer.

3) Air-Ground Channel Model.

The channel from the SIM to eVTOL m is modeled as a Rician fading channel. The k -th entry is:

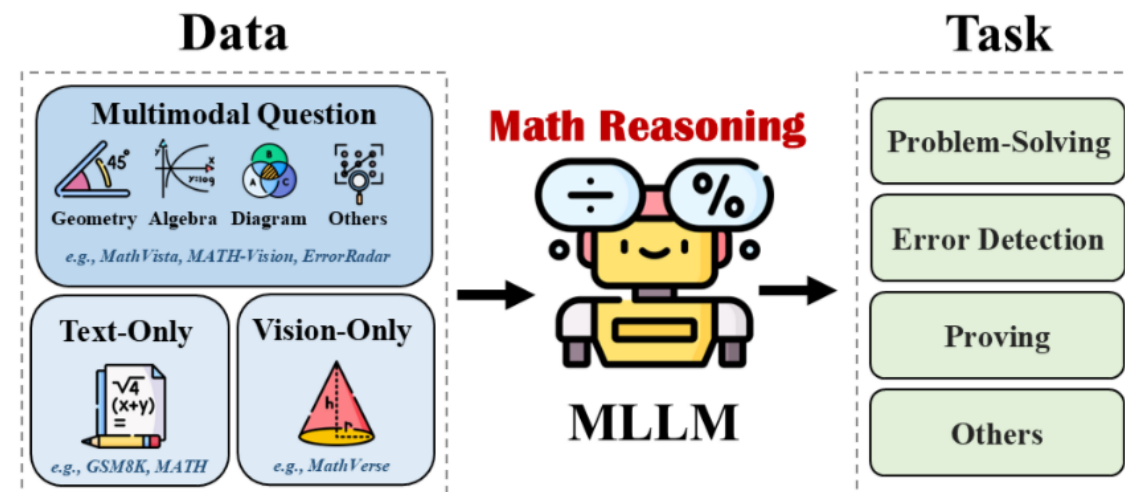
$$h_{m,k}[n] = \sqrt{\frac{\rho_0}{(d_m[n])^{\alpha^h}}} \sqrt{\frac{\kappa^h}{\kappa^h + 1}} \tilde{h}_m[n]$$

无线大语言模型

• 模型微调 (Mathematical Reasoning)

- 群体相对策略优化 (GRPO)
 - DeepSeek模型中做Mathematical Reasoning的方法
- 基于开源的小参数大语言模型
 - 通义千问模型 (Qwen2.5 [0.5B, 3B, 7B])
- 基于强化学习的方法
 - 目标函数如下:

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E}_{\substack{x \sim P(X) \\ \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(\frac{\pi_{\theta}(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} A_i, \text{clip} \left(\frac{\pi_{\theta}(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)}, 1 - \epsilon, 1 + \epsilon \right) A_i \right) \right]$$



[3] Yan, Yibo, et al. "A survey of mathematical reasoning in the era of multimodal large language model: Benchmark, method & challenges." arXiv preprint arXiv:2412.11936 (2024).

[4] Shao, Zhihong, et al. "Deepseekmath: Pushing the limits of mathematical reasoning in open language models." arXiv preprint arXiv:2402.03300 (2024).

无线大语言模型

• 问答样例

Background

In a wireless semantic communication system with interference, the received real-valued signal after equalization combines the desired signal, an interference signal, and noise. The system model is derived from the complex baseband representation, where $\mathbf{y} \in \mathbb{R}^{2k}$ is the equalized received real signal vector, $\mathbf{x} \in \mathbb{R}^{2k}$ is the real-valued semantic feature vector to be transmitted, $\mathbf{z} \in \mathbb{R}^{2k}$ is the real-valued interference vector, $\mathbf{n} \sim \mathcal{N}(0, \frac{\sigma^2}{2} \mathbf{I}_{2k})$ is the real-valued additive white Gaussian noise vector, $P_x \in \mathbb{R}^+$ is the desired signal transmit power (in linear scale), $P_z \in \mathbb{R}^+$ is the interference signal transmit power (in linear scale), $\mathbf{W}_x \in \mathbb{R}^{2k \times 2k}$ is the channel transformation matrix for the desired signal, $\mathbf{W}_z \in \mathbb{R}^{2k \times 2k}$ is the channel transformation matrix for the interference signal, and $\mathbf{W}_n \in \mathbb{R}^{2k \times 2k}$ is the channel transformation matrix for the noise.

Question

Write the complete received signal equation including all three components.

Answer: $\sqrt{P_x} \mathbf{W}_x \mathbf{x} + \sqrt{P_z} \mathbf{W}_z \mathbf{z} + \mathbf{W}_n \mathbf{n}$

背景介绍

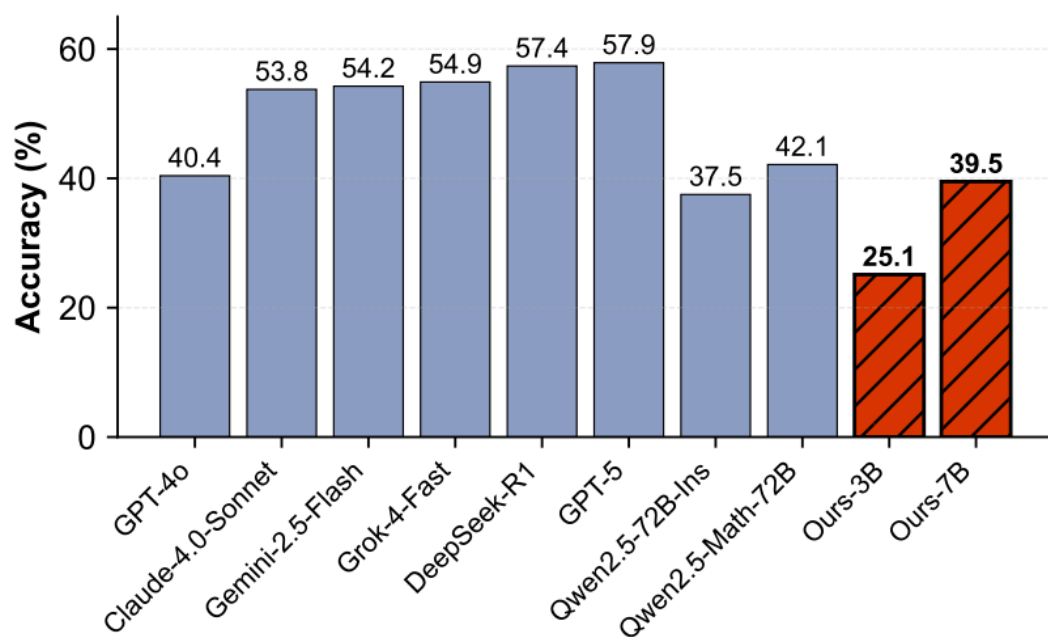
提问

回答

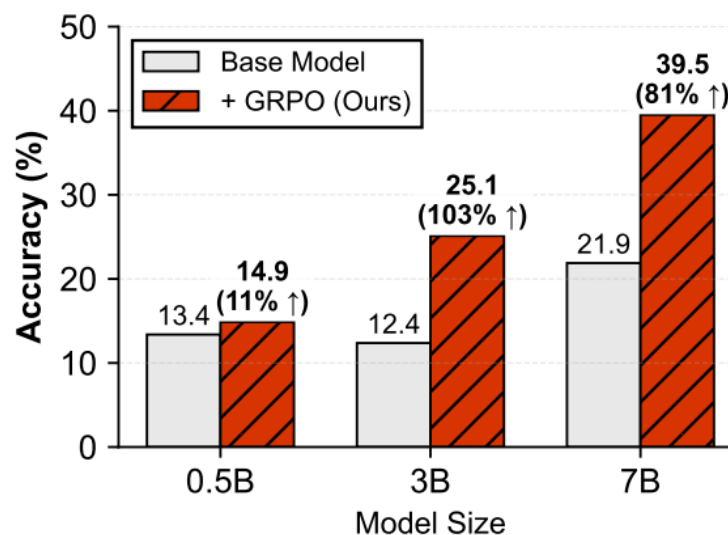


无线大语言模型

• 实验结果



- 参数量小的大语言模型 + 专业知识库训练
- 参数量更大的通用大语言模型
- 相似水平的逻辑表现



- 强化学习 (GRPO) + 高质量专业数据集
- 大幅度强化大语言模型在专业领域的问答表现

无线大语言模型

• 总结

- 很早期的无线领域大语言模型LLM工作
- 不仅仅是在prompt层面做改进
- 建立了专业的无线领域数据集 + 微调

Model	MCQ	Progressive Masking Filling			FEC	Avg. Acc
		Level 1	Level 2	Level 3		
DeepSeek-R1	76.00%	60.00%	34.91%	12.50%	7.83%	38.05%
OpenAI-o1	66.40%	59.17%	32.17%	8.04%	6.96%	34.55%
GPT-4o	72.80%	42.50%	28.70%	6.25%	4.35%	30.92%
GPT-4o (w cot)	72.00%	40.00%	23.48%	4.46%	5.22%	29.03%
GPT-4	53.60%	38.33%	18.26%	3.57%	4.35%	23.62%
GPT-4 (w cot)	58.40%	32.50%	14.78%	3.57%	4.35%	22.72%
GPT-3.5-turbo	45.60%	7.50%	10.43%	1.79%	1.74%	13.41%
GPT-3.5-turbo (w cot)	48.80%	11.67%	8.70%	2.68%	2.61%	14.89%
DeepSeek-V3	78.40%	50.00%	24.35%	6.25%	6.96%	33.19%
DeepSeek-V3 (w cot)	73.60%	50.00%	20.00%	6.25%	6.09%	31.19%
Gemini-2.0-flash	71.20%	40.83%	24.35%	5.36%	4.35%	29.22%
Gemini-2.0-flash (w cot)	73.60%	40.00%	24.35%	8.04%	4.35%	30.07%
Gemini-1.5-pro	65.60%	43.33%	29.57%	9.82%	6.09%	30.88%
Gemini-1.5-pro (w cot)	69.60%	38.33%	25.22%	7.14%	3.48%	28.75%
Gemini-1.5-flash	66.40%	37.50%	13.91%	2.68%	4.35%	24.97%
Gemini-1.5-flash (w cot)	68.80%	32.50%	17.39%	2.68%	5.22%	25.32%

? 哪个通用大语言模型在无线领域更专业

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